Input

Client : Hello

Server : Hi

Output

Client : Hello

Server : Hi.

Ex: 12 (b)

CHAT CLIENT- SERVER USING TCP SOCKETS.

AIM:

To implement a simple chat client - server application using TCP sockets.

ALGORITHM:

SERVER

1. Create a TCP socket.
2. Bind and listen for a client.
3. Accept the connection.
4. Receive a message from the client.
5. Send a reply to the client.
6. Close the connection.

CLIENT:

1. Create a TCP socket.
2. Connect to the server.
3. Send a message.
4. Receive the server's reply.
5. Display the reply.
6. Close the connection.

CODE - SERVER

```
import socket
s = socket.socket()
s.bind("localhost", 5000)
s.listen()
c, addr = s.accept()
while True:
    msg = c.recv(1024).decode()
    if msg == "bye": break
    print("client", msg)
    c.send(input("server:").encode())
c.close()
s.close().
```

CODE - CLIENT

```
import socket
s = socket.socket()
s = connect(("localhost", 5000))
while True:
    msg = input("client:")
    s.send(msg.encode())
    if msg == "bye": break.
    print("server:", s.recv
          (1024.decode())
s.close()
```

RESULT:

Thus the chat client